

Tests & Quizzes

Quiz 2

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Part 1 of 1 1.0 / 5.0 Points

Question 1 of 5

0.0

1.0 Points

Why can neither Proof of Work nor Proof of Stake guarantee consensus if a majority of influence becomes malicious?

- ✗ A. Because cryptographic verification becomes unreliable beyond that point.
- ✓ B. Because block validation becomes computationally infeasible.
- ✓ C. Because peer-to-peer networking protocols collapse.
- ✓ D. Because the group having majority of influence has the power to determine the accepted chain.

Answer Key: D

Question 2 of 5

0.0

1.0 Points

In the context of the Byzantine Generals Problem, which of the following situations does NOT represent a general acting maliciously or deceitfully?

- ✓ A. A general sending different battle plans to different generals.
- ✗ B. A general issuing misleading instructions to create confusion.
- ✓ C. A general transmitting orders after a significant communication delay.
- ✓ D. A general deliberately providing false commands while appearing cooperative.

Answer Key: C

Question 3 of 5

1.0

1.0 Points

Consider the following sequence in a blockchain transaction lifecycle:

1. A transaction is created and broadcast to the network.
2. The transaction is validated and grouped with other transactions.
3. Network participants reach agreement on a proposed block, establishing a new authoritative state.
4. The agreed block is propagated and stored locally by all nodes.

At which step is the ledger logically updated?

- ✓ A. Step 1.
- ✓ B. Step 4.
- ✓ C. Step 3.
- ✓ D. Step 2.

Answer Key: C

Question 4 of 5

0.0

1.0 Points

You are validating transactions and constructing a block. Before you broadcast your block, you receive a valid block mined by another participant. That block has been broadcast but is not yet finalized by the network. What is the **most appropriate action** you should take?

- ✓ A. Discard your block and begin validating on top of the received block.
- ✓ B. Pause your current work and resume validation only after the received block is finalized.
- ✓ C. Continue mining your block until a fork is explicitly detected.
- ✗ D. Immediately broadcast your own block to compete for acceptance.

Answer Key: A

Question 5 of 5

0.0

1.0 Points

A system has **multiple writers**, and all writers are **known and trusted entities**. Which design choice is most appropriate?

- ✓ A. Public permissioned blockchain.
- ✗ B. Private permissioned blockchain.
- ✓ C. Do not use a blockchain.
- ✓ D. Public permissionless blockchain.

Answer Key: C

